

## Differential equations:

In calculus, you learned about functions, their derivatives, and their integrals. The basic idea of differential equations is to specify the function,  $x(t)$ , through its derivatives and initial conditions.

$$\underbrace{\frac{dx}{dt} = f(x, t)}_{\text{core equation}}, \quad \underbrace{x(0) = x_0}_{\text{initial conditions}}$$

Since the above depends only on first-order derivatives, it is called a first order differential equation.

Simplest case:

$$\frac{dx}{dt} = f(t) \Rightarrow \int_0^t dt' \frac{dx}{dt'} = \int_0^t dt' f(t')$$

$$\Rightarrow x(t) - x(0) = \int_0^t dt' f(t')$$

$$\Rightarrow x(t) = x_0 + \int_0^t dt' f(t')$$

Typically the righthand side will depend on  $x$ , and solutions require more work.

Does anyone recognize these canonical differential equations from calculus?

$$(i) \quad \frac{dx}{dt} = x, \quad x(0) = 1$$

$$(ii) \quad \underbrace{\frac{d^2x}{dt^2} = -x}_{\text{2nd-order}}, \quad \underbrace{x(0) = 0, \frac{dx}{dt}(0) = 1}_{\text{initial conditions now involve } dx/dt}$$

The solutions are,

$$(i) x(t) = e^t$$

$$(ii) x(t) = \sin(t)$$

as can be verified by taking their derivatives.

Method of separation: Suppose that

$$\frac{dx}{dt} = f(x)g(t)$$

Since the righthand side separates as a product, it can be solved by rearrangement and integration.

$$\frac{dx}{f(x)} = g(t)dt \Rightarrow \int_{x_0}^{x(t)} \frac{dx}{f(x)} = \int_0^t dt' g(t')$$

This allows us to solve (i).

$$\begin{aligned} \frac{dx}{dt} = x &\Rightarrow \int_{x_0}^{x(t)} \frac{dx}{x} = \int_0^t dt' \Rightarrow \underbrace{\ln(x)}_{\text{natural logarithm}} \Big|_{x_0}^{x(t)} = t \\ &\Rightarrow \underbrace{\ln\left(\frac{x(t)}{x_0}\right)}_{\ln(a) - \ln(b) = \ln\left(\frac{a}{b}\right)} = t \Rightarrow x(t) = x_0 e^t \end{aligned}$$

In neuroscience, we'll often encounter the equation

$$\tau \frac{dx}{dt} = -x + I$$

This can be recast in the exponential form when  $I$  does not depend on  $t$ . Change variables

$$t' = -t/\tau \Rightarrow dt = -\tau dt'$$

$$x' = -x + I \Rightarrow dx = -dx'$$

In these variables

$$\frac{dx'}{dt'} = x' \Rightarrow x'(t') = x'(0) e^{t'}$$

Going back to the original variables

$$\underbrace{-X(t) + I}_{\text{Gap between } X \text{ and } I} = \underbrace{(-X_0 + I)}_{\text{Initial gap}} \underbrace{e^{-t/\tau}}_{\text{Exponential decay with timescale } \tau}$$

$$X(t) = \underbrace{X_0 e^{-t/\tau}}_{\text{Decay}} + \underbrace{(1 - e^{-t/\tau}) I}_{\text{Exponential approach}}$$

Euler's method:

Let's return to the general first-order equation

$$\frac{dx}{dt} = f(x, t), \quad x(0) = X_0$$

For most choices of  $f$ , we can't solve the equation by hand, and we need to use a computer. Euler proposed the simplest way to do this. From calculus

$$\frac{dx}{dt} = \lim_{\Delta t \rightarrow 0} \frac{x(t + \Delta t) - x(t)}{\Delta t}$$

Euler's method fixes  $\Delta t$  to a small value and uses the righthand side instead of the derivative

$$\frac{x(t + \Delta t) - x(t)}{\Delta t} = f(x(t), t)$$

$$x(t + \Delta t) = x(t) + \Delta t f(x(t), t)$$

This update equation can be run repeatedly to simulate the system.

MATLAB demonstration of Euler's method:

See code

Higher-order differential equations: Can be rewritten as a set of coupled first-order equations. Let's illustrate with the example (ii).

$$\frac{d^2 x}{dt^2} = -x, \quad x(0) = 0, \quad \frac{dx}{dt}(0) = 1$$

Introduce a new equation and variable

$$y = \frac{dx}{dt} \Rightarrow \frac{dy}{dt} = \frac{d^2 x}{dt^2} = -x$$

$$\Rightarrow y(0) = 1$$

The second-order equation is thus equivalent to:

$$\frac{dx}{dt} = y, \quad x(0) = 0$$

$$\frac{dy}{dt} = -x, \quad y(0) = 1$$

Later in the course, we'll solve equations like these by hand. For now we'll simulate them.